Skills Summary

Vertices, Foci and Asymptotes

Use this reference while completing your worksheet or when studying.

Skill 1 — Ellipse axes and vertices

Read the denominators. In $\dfrac{x^2}{a^2}+\dfrac{y^2}{b^2}=1$ the denominators are the *squares* of the half-axes, so take a square root before reporting a length.

Which axis is long? The larger denominator names it: larger under x^2 means a horizontal major axis. Full axis lengths are $2a$ and $2b$ — twice the half-lengths, which is what a physical width or height usually means.

Example: For $\dfrac{x^2}{49}+\dfrac{y^2}{25}=1$, find both axis lengths.
Step 1. The question asks for widths, not half-widths, so square-root each denominator and then double — the doubling is where most marks are lost.
Step 2. $\sqrt{49}=7$ and $\sqrt{25}=5$, so the half-axes are 7 and 5.
$\Rightarrow$ major axis **14**, minor axis **10**

Try it: For $\dfrac{x^2}{81}+\dfrac{y^2}{36}=1$, find both axis lengths.

check: major **18**, minor **12**

Skill 2 — Ellipse foci

Subtract: $c^2=a^2-b^2$, where a is the larger half-axis. The foci lie on the long axis, at $\pm c$ from the centre, so they are always *inside* the curve.

The defining property: the two distances from any point of the ellipse to the two foci always add to $2a$. That is why a whispering gallery works, and it is often what an application is really asking for.

Example: For $\dfrac{x^2}{100}+\dfrac{y^2}{64}=1$, find the focal distance and the separation of the foci.
Step 1. Foci sit inside an ellipse, so the rule is the subtracting one — and the question's "separation" then needs the doubling as well.
Step 2. $c=\sqrt{100-64}=\sqrt{36}$, and the two foci are $2c$ apart.
$\Rightarrow$ $c=\mathbf{6}$, separation **12**

Try it: For $\dfrac{x^2}{169}+\dfrac{y^2}{25}=1$, find the focal distance and the separation of the foci.

check: $c=\mathbf{12}$, separation **24**

Skill 3 — Hyperbola vertices and foci

The positive term decides everything. If x^2 is positive the branches open left and right; if y^2 is positive they open up and down. The vertices sit at $\pm a$ from the centre, where a^2 is the denominator of the *positive* term.

Add: $c^2 = a^2 + b^2$ — the opposite of the ellipse rule, and the reason a hyperbola's foci always lie beyond its vertices.

Example: For $\dfrac{x^2}{16} - \dfrac{y^2}{9} = 1$, find the vertex and focal distances.

Step 1. The x^2 term is the positive one, so it supplies a; after that the only choice is add-or-subtract, and hyperbolas add.

Step 2. $a = \sqrt{16} = 4$, and $c = \sqrt{16+9} = \sqrt{25}$.

$\Rightarrow$ vertex at **4**, focus at **5** (focus beyond vertex, as always)

Try it: For $\dfrac{y^2}{36} - \dfrac{x^2}{64} = 1$, find the vertex and focal distances.

check: vertex **6**, focus **10**

Skill 4 — Hyperbola asymptotes

Slopes depend on orientation. For $\dfrac{x^2}{a^2} - \dfrac{y^2}{b^2} = 1$ the asymptotes are $y = \pm\dfrac{b}{a}x$; for $\dfrac{y^2}{a^2} - \dfrac{x^2}{b^2} = 1$ they are $y = \pm\dfrac{a}{b}x$. In both cases the number over the *positive* term goes on top.

What they mean: far from the centre the branches get arbitrarily close to these lines and never touch them.

Example: Find the asymptote slopes of $\dfrac{x^2}{25} - \dfrac{y^2}{9} = 1$.

Step 1. The positive term is the one in x, which puts the y-denominator on top of the slope fraction — get the orientation right before touching the arithmetic.

Step 2. $a = \sqrt{25} = 5$ and $b = \sqrt{9} = 3$, so the slopes are $\pm b/a$.

$\Rightarrow$ slopes $\pm\boldsymbol{\frac{3}{5}}$

Try it: Find the asymptote slopes of $\dfrac{y^2}{16} - \dfrac{x^2}{49} = 1$.

check: $\pm\boldsymbol{\frac{4}{7}}$